

Homework #4

Due date : **June 19th, 2026**

1. Please submit your report (*.pdf) and schematic file (*.asc) to EECLASS [-5 pt]
2. Please name your report as hw3_studentID.pdf (e.g. hw3_113061000.pdf) [-5 pt]
3. Please name your schematic file as hw3_studentID.asc. If you submit additional files for specific problems, please name them as hw3_studentID_xxx.asc, where xxx indicates the problem name. For example: hw3_113061000_c.asc and hw3_113061000_d.asc. [-5 pt]
4. Please set white background and data trace with width = 3 in tutorial. [-5 pt]
5. You MUST extract the data of the waveform by cursor in tutorial. [-5 pt]
6. You MUST attach screenshot of the schematic with simulation commands and output waveform in your report. [-5 pt]
7. List calculation/design process clearly and mark proper information of all plots/figure required. [-5 pt]
8. Screenshots of the waveforms must include both x-axis and y-axis labels. [-5 pt]

Design of a MOS amplifier

Consider the Common-Source amplifier in Fig.1, where $V_{DD} = 3V$, $R_{sig} = 3k\Omega$, $R_L = 10k\Omega$, and $C_{CI} = C_{CO} = C_S = 5\mu F$. **Consider the use of $0.5\mu m$ CMOS technology for the MOS and use the SPICE level-1 model in Table 1.** Please follow the steps below to **design R_{G1} , R_{G2} , R_D , and R_S** to achieve a voltage gain of $|v_o/v_{sig}| \geq 30$, and a power consumption less than **$1mW$** (Hint: Use **“.model <model_name> NMOS(level=1 tox=9.5e-9 ...)”** for your transistors. **Make sure the model name in SPICE directive matches the model names in schematic.**)

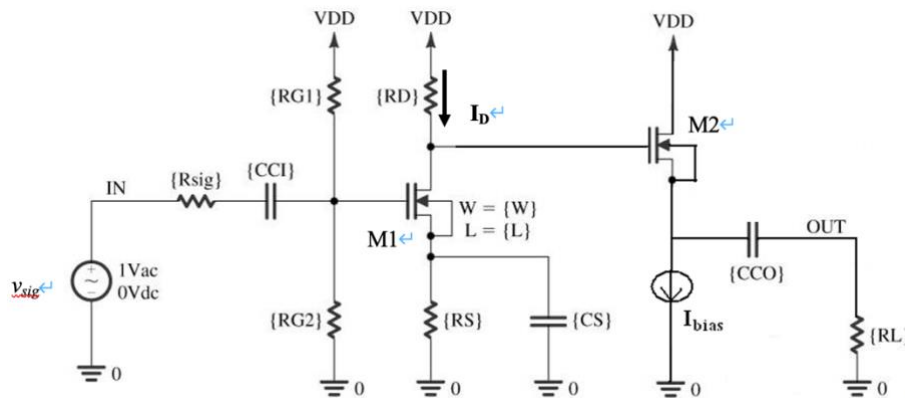


Fig. 1

Table 1:

	5- μm CMOS Process		0.5- μm CMOS Process	
	NMOS	PMOS	NMOS	PMOS
LEVEL	1	1	1	1
TOX	85e-9	85e-9	9.5e-9	9.5e-9
UO	750	250	460	115
LAMBDA	0.01	0.03	0.1	0.2
GAMMA	1.4	0.65	0.5	0.45
VTO	1	-1	0.7	-0.8
PHI	0.7	0.65	0.8	0.75
LD	0.7e-6	0.6e-6	0.08e-6	0.09e-6
JS	1e-6	1e-6	10e-9	5e-9
CJ	0.4e-3	0.18e-3	0.57e-3	0.93e-3
MJ	0.5	0.5	0.5	0.5
CJSW	0.8e-9	0.6e-9	0.12e-9	0.17e-9
MJSW	0.5	0.5	0.4	0.35
PB	0.7	0.7	0.9	0.9
CGBO	0.2e-9	0.2e-9	0.38e-9	0.38e-9
CGDO	0.4e-9	0.4e-9	0.4e-9	0.35e-9
CGSO	0.4e-9	0.4e-9	0.4e-9	0.35e-9

$$C_{ox} = \frac{3.9\epsilon_0}{t_{ox}}$$

a.) **Design the bias current I_D , I_{bias} , W/L and R .** Follow the procedures below:

1. Design biasing current I_D and I_{bias} to satisfy the power specification.
2. Choose $V_{RD}=V_{DS1}=V_{DD}/3$ (A typical value for low voltage design). Design R_D and R_S .
3. Design the W/L ratio of M1 to satisfy the gain requirement.
4. Design the W/L ratio of M2 to satisfy the DC biasing condition chosen in (2). Check whether $(1/gm_2) \ll R_L$? If not, return to step (1) and choose a new I_{bias} and I_D
- 5 . Design all other resistors to bias the MOS with the desired bias current derived in (1).
(Hint : $V_G=V_{DD}\frac{R_{G2}}{R_{G1}+R_{G2}}$, $V_S=I_D R_S$, $V_{D1}=V_{DD} - I_D R_D$)

b.)**Frequency Response.** Let v_{sig} be a 1-V ac voltage source. Sweep the frequency response of the amplifier and **plot the gain response** to check whether the midband gain of the amplifier satisfied the requirement. **Mark and note the values of the lower and higher 3-dB frequency explicitly in your plot.**

c.)**Transient Simulation.** Use **hand calculation** to derive the dynamic ranges for v_o and the corresponding dynamic range for v_{sig} . Use transient simulation in Pspice to verify your calculation and **plot the v_o and v_{sig} versus time.** (Hint : the frequency of the sinusoidal signal v_{sig} should be chosen to lie in the midband frequency range concluded in (b))

d.) **Influence of transistor M2.** Remove the transistor M2 and connect the output load (R_L & CCO) to the drain of M1. Then sweep the frequency response again. **Plot the new gain response, identify the differences** from the results in (b), and **explain these differences quantitatively** as possible.

e.) **Influence of C_s .** Modify the value of C_s to be zero and sweep the frequency response again. **Plot the new gain response, identify the differences** from the results in (b), and **explain these differences quantitatively** as possible.

f.) **Influence of C_{gd} .** Add a capacitor of 0.5nF between the drain and the gate of the MOS. Sweep the frequency response again. **Plot the new gain response, identify the differences** from the results in (b), and **explain these differences quantitatively** as possible.